

Beyond Core GRADE. Risk of bias in individual studies: Randomised trials: Alternative approach to addressing missing outcome data

To decide whether to rate down certainty of evidence for risk of bias in randomized control trials, Core GRADE suggests systematic reviewers use one of the available instruments to assess risk of bias associated with each item for each included study. Ultimately, these ratings inform judgments of the overall risk of bias of each individual study as low or high.

For missing outcome data, one instrument recommended for Core GRADE, ROBUST-RCT,¹ identifies two possible approaches. One is simple: set thresholds for percent missing data for each rating of risk of bias (e.g. missing data < 5% low risk of bias, 5% to 10% probably low, 10% to 15% probably high, over 15% high risk of bias). A second more sophisticated approach involves systematic reviewers conducting a primary meta-analysis using complete-case analysis (i.e., include only the participants whose outcome data are available) followed by sensitivity meta-analyses imputing, for each individual study, data for the participants whose outcome data are unavailable.

The aim of the approach is to test the robustness of the inference made by the primary meta-analysis. For example, for an analysis of a binary outcome, if a primary meta-analysis suggests a beneficial effect of an intervention (i.e., the intervention decreases risk of an undesirable outcome), reviewers may assume the event rates in the comparator group in those followed successfully and in those with missing data are the same. In contrast, the sensitivity analyses will challenge the robustness of the inference of beneficial intervention effect by assuming in the intervention group the event rate in the participants with missing data is higher than the event rate in those whose outcome data are available. The ratio of the two event rates may range from 1.5 (the least challenging assumption) to 5 (the most challenging assumption).² Systematic reviewers need to decide which strategy represents the most extreme but still plausible in their specific context.

For continuous outcomes, reviewers need to impute both means and standard deviations for participants with missing data in each group. If a primary meta-analysis suggests a beneficial effect of intervention, the serial imputations attribute progressively worse outcomes to those in the intervention group with missing data, and progressively better outcomes in comparator group with missing data. The sensitivity analyses thus provide progressively more stringent criteria for inferring robustness of the results.² Once again, systematic reviewers need to decide which strategy represents the most extreme but still plausible in their specific context. Systematic reviewers can adopt similar approach to addressing risk of bias related to missing data for time-to-event outcomes.³

The ultimate goal of this approach is to decide whether to rate down certainty of body of evidence for missing outcome data. If the inference is robust to the most extreme assumption viewed as plausible, one does not need to rate down. Otherwise, rating down is warranted.

This approach addresses risk of bias associated with missing data across the entire body of evidence at the meta-analysis level rather than at individual study level. As described above, it is more sophisticated and not so familiar to systematic reviewers. Thus, we consider this approach as an alternative beyond Core GRADE.

References

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3. Goldkuhle M, Bender R, Akl EA, et al. GRADE Guidelines: 29. Rating the certainty in time-to-event outcomes-Study limitations due to censoring of participants with missing data in intervention studies. *J Clin Epidemiol* 2021;129:126-37. doi: 10.1016/j.jclinepi.2020.09.017 [published Online First: 20200930]